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Extractive desulfurization of model jet fuel using metal chloride modified deep eutectic solvents

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ABSTRACT

Crude oil contains various organic sulfur compounds comprising of heterocyclic and paraffinic hydrocarbons. During upgradation of crude oil, these sulfur compounds inevitably end up in the final products, which include different fuel fractions (gasoline, kerosene, diesel, etc.). Jet fuels are one such crude oil fraction. Combustion of high sulfur containing fuels lead to sulfur oxide emissions which is detrimental for the environment. Hydrodesulfurization (HDS) process is the conventional fuel desulfurization approach. However, this method is cost and energy intensive therefore, alternative desulfurization approaches are continually being investigated. One such fuel desulfurization route is solvent extraction utilizing deep eutectic solvents (DESs). In this investigation, a choline chloride and glycerol based DES has been formulated. Four different metal chlorides viz. iron(III) chloride, copper(II) chloride, zinc(II) chloride and tin(II) chloride were added to the DES with an objective to enhance the desulfurization efficiency. These solvents were characterized by FTIR, rheology, surface tension and Kamlet-Taft parameters. The effectiveness of these solvents towards desulfurization of model jet fuel has been analysed. A molecular dynamic simulation of the solvents as well as the model fuel has also been carried out for better understanding of the desulfurization process. Both the experimental and simulated findings indicated that the desulfurization efficiency of the DES containing copper(II) chloride was the maximum. The highest desulfurization efficiency of 99.15 % (on total sulfur basis) with a solvent:feed ratio of 1:5 at an operating temperature of 25 °C and an extraction time of 3 h was achieved using the aforementioned DES.

1. Introduction

Organic sulfur compounds are an integral part of petroleum distillates, including fuels. Jet fuel is no exception and untreated jet fuel can contain as high as 10,000 ppm of sulfur. Sulfur laden fuels when combusted in engines, lead to the emission of sulfur oxides (SO_x), which eventually causes environmental pollution. Some of the undesirable effects of SO_x pollution include acid rain, formation of obnoxious smog which can cause severe respiratory and cardiovascular problems in humans [1]. This has necessitated stringent measures from regulatory bodies all over the world to reduce the sulfur content in fuels, below a certain threshold [2].In the case of jet fuels, the maximum permissible limit of sulfur is 3000 ppm since most of the emissions from jet fuel occur at high altitudes which have low impact on the environment. However, some of the jet emissions at lower altitudes, especially near airports could have detrimental effects. Also, this high sulfur content in jet fuels make them unsuitable for alternative applications viz. in fuel cells [3]. Hydrodesulfurization (HDS) utilizing NiMo/CoMo catalysts is the stereotypical fuel desulfurization approach in refineries [4]. However, this process operates under high pressure and high temperature and utilizes expensive proprietary catalysts making itboth energy and cost intensive. Also, HDS process is not very efficient in removing heterocyclic sulfur compounds under standard operating conditions due to low reactivity of these compounds and higher severity (i.e. higher temperature and pressure) is required to remove these compounds [1,2,4,5]. Some of the alternative desulfurization approaches which have been previously investigated include oxidative desulfurization, extractive desulfurization, adsorptive desulfurization and biodesulfurization. One common advantage of these processes over the conventional HDS is that these can be conducted at comparatively less severe conditions [1,5–7].

Desulfurization via solvent extraction is a preferable alternative since this process does not usually require additional reagents viz. hydrogen or oxidizing agents thereby reducing the operating cost; the solvents can be regenerated and recycled and additionally the properties of the fuel mixture mostly remain unaltered [2,5,7].

Conventional solvents viz. *N,N*-dimethylformamide, dimethyl

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sulfoxide, imidazolidinone, *N*-methylpyrrolidone, methanol, acetonitrile, sulfolane, dimethylformamide and dimethyl sulfoxide can be utilized for this purpose, but the toxicity and environmental hazards posed by these volatile organic solvents make them unfit for a green process. Thus, it is necessary to explore the possibilities of utilizing ecofriendly solvents for extractive desulfurization process featuring good extraction capacity, high selectivity towards the sulfur compounds, easy regeneration, low volatility and viscosity, and high thermal stability. Two emerging classes of environment friendly solvents which have exhibited fuel desulfurization capacities are ionic liquids (ILs) and deep eutectic solvents (DESs). ILs satisfy the criteria of good ecofriendly solvents and their highly tunable nature are suitable for extractive fuel desulfurization [2,8]. ILs though a class of green solvents, possess certain drawbacks viz. high raw material cost and complex synthesis and purification steps involved in their production. DESs overcomes these drawbacks since these are easy to formulate, utilizes cheaper raw materials and there are no byproducts formed thus eliminating the need for further purification. DESs are formed by simply mixing a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD) at a specific molar ratio. The freezing point of the resulting mixture is much lower than the freezing point of either the HBA or HBD. There are several HBAs and HBDs available therefore numerous DESs having variable properties can be prepared. Thus, it is important to select the HBA and HBD precisely to suit a specific application [4,6,8]. The selection of DES components can be accomplished either by reports published in literature pertaining to the specific application or through machine learning tools like Conditional Generative Adversarial Network (CGAN), machine learning models utilizing multilayer perceptron and support vector regression [9,10].

Some applications of DESs are as a green reaction media, electrolyte in electrochemistry, porous material synthesis media, as catalyst and as a solvent in different extraction processes [1,5]. In recent years, DESs have seen application in crude oil derived fuel processing including dearomatization, denitrogenation and desulfurization [5].

The tetrapropyl ammonium bromide and acetic acid based deep eutectic solvent was utilized by Tarek Lemaoui et. al., for simultaneous dearomatization, desulfurization and denitrogenation of model diesel, attaining efficiencies of approximately100%, 68 % and 89 % respectively [11].

In another investigation conducted by Sajith Babu M.K. et. al., a similar integrated impurity removal from model diesel was performed using a DES consisting of imidazole and sesamol. Extraction efficiencies of nearly100%, 71.1 % and 56.1 % for desulfurization, dearomatization and denitrogenation respectively after single treatment were achieved [12].

In this investigation, the selection of HBA and HBD was accomplished based on the review of literature. Among the different HBAs available, choline chloride stands out because it is a class of quaternary ammonium salt (which has been utilized for DES formulation in prior investigations), is readily available, cheap and biodegradable [13–15]. Choline chloride as the HBA also provides the integral melting point depression, which is required for the formation of a DES, due to its low enthalpy of fusion (4.3 KJ/mol). The lower intermolecular interaction tendency of ChCl also leads to lower viscosity of DES which is expected to facilitate higher mass transfer between phases in liquid-liquid extraction [16,17].

Glycerol a non-toxic byproduct from biodiesel production, is a suitable HBD for a quaternary ammonium salt based DES since it can form good hydrogen bond with choline chloride [1,17,18]. Additionally, when glycerol is used as the HBD, it inhibits the hydrogen bond interaction among the glycerol molecules, resulting in overall decrease in viscosity of the DES. The investigation conducted by the National Institute of Standards and Technology of the United States showed that when ChCl and glycerol were used as the HBA and HBD respectively, the choline ions entered the hydrogen bond network gaps between the glycerol molecules resulting in “isolation effect” thereby reducing the viscosity of the system. Reduction in viscosity leads to efficient interfacial mass transfer [19].

The HBA (choline chloride) and HBD (glycerol) selected for the present investigation forms DES when mixed in the molar ratio of 1:2, as also reported by earlier investigators [4].

The desulfurization efficiency of a DES can be enhanced by incorporating a metal chloride, as a results of π-complex formation between the empty orbital of metals and the un-bonded electron pairs of sulfur [1,20,21].

The work pursued by Zhang, Li.C., et al., indicated that the desulfurization performance for model fuel utilizing metal chlorides based DES followed the order $FeCl_3 > ZnCl_2 > CuCl_2$ which was attributed to the size of electron cloud distribution of the respective metal ions. The Fe^{3+} ion having the largest electron cloud, could interact readily with the sulfur compounds resulting in the best performance [20].

In another investigation by Shirazinia, S.R., et al., $SnCl_2$, $CuCl_2$, $FeCl_3$, $ZnCl_2$ based DESs were used for extractive desulfurization of fuel. The desulfurization efficiencies of the metal chlorides followed the order $SnCl_2 > CuCl_2 > FeCl_3 > ZnCl_2$. The highest desulfurization performance of $SnCl_2$ based DES was ascribed to Sn, which being a soft post transition metal does not form a strong complex with the components of DES, thereby can interact more readily with the sulfur compounds giving rise to a high desulfurization efficiency [1].

In this investigation, a DES comprised of choline chloride and glycerol has been formulated and modified with four different metal chlorides viz. iron(III) chloride, copper(II) chloride, zinc(II) chloride and tin (II) chloride to form novel metal chloride based DESs (MCDESs), with an objective to desulfurize model jet fuel.The MCDESs have been characterized by Fourier transform infrared spectroscopy (FTIR), rheology and interfacial surface tension. The Kamlet-Taft (K-T) parameters of the MCDESs were measured and reported for the first time to predict the desulfurization behaviour of the solvents. The effect of different metal chlorides and their varying concentrations in the synthesized DESs on the desulfurization efficiency of model jet fuel was explored to choose the best MCDES. Desulfurization of jet fuel utilizing MCDESs has not been reported by earlier investigators. The effect of different operating parameters on extractive desulfurization was optimized using Analysis of Variance (ANOVA). Additionally, geometry optimization using Materials Studio software version 5.5.2 was also performed for the first time, to predict the affinity of each of the components of the MCDESs towards the sulfur compounds in the model jet fuel desulfurization.

2. Materials and methods

2.1. Materials

Choline chloride (≥98 %) (ChCl), glycerol (≥98 %), ferric chloride (98 %) ($FeCl_3$), copper chloride (98 %) ($CuCl_2$), tin chloride (≥97 %) ($SnCl_2$) and zinc chloride (97 %) ($ZnCl_2$) were utilized for the preparation of MCDESs. n-decane(≥99 %), tolune (99 %), n-butane thiol (99.99 %) (thiol) and 1-benzothiophene (≥99 %) (BT) were used to formulate the model jet fuel. N,N-diethyl-4nitro aniline, 4-nitro aniline and Nile red dyes were employed for the determination of Kamlet-Taft parameters. Mohr’s salt (≥83 %) ($(NH_4)_2Fe(SO_4)_2{\cdot}6H_2O$), potassium dichromate (≥96 %) ($K_2Cr_2O_7$), ferroin indicator, mercury sulphate (≥98 %) ($HgSO_4$), silver sulphate (≥99 %) (Ag_2SO_4), sulphuric acid (≥98 %) (H_2SO_4), manganese sulphate monohydrate (≥87 %) ($MnSO_4.H_2O$), sodium azide (≥98 %) (NaN_3), potassium iodide (≥98 %) (KI), potassium hydroxide (≥87 %) (KOH), phosphoric acid (H_3PO_4), magnesium sulphate (≥98 %) ($MgSO_4$), calcium chloride (≥99 %) ($CaCl_2$) were used to evaluate the biodegradability of MCDES. All the reagents were procured from Merck Specialities Pvt. Ltd., and were used as received.

Table 1
Composition of the formulated MCDESs.

Molar ratio of choline chloride: glycerol: metal chloride	Metal chloride used	Nature of the metal chloride incorporated DES	Abbreviation of fresh MCDESs	Abbreviation for used MCDESs
1:2:0.05	$FeCl_3$	Homogenous mixture	FeDES0.05	FeDES0.05 U
1:2:0.20	$FeCl_3$	Homogenous mixture	FeDES0.20	FeDES0.20 U
1:2:0.35	$FeCl_3$	Homogenous mixture	FeDES0.35	FeDES0.35 U
1:2:0.50	$FeCl_3$	Homogenous mixture	FeDES0.50	FeDES0.50 U
1:2:0.05	$CuCl_2$	Homogenous mixture	CuDES0.05	CuDES0.05 U
1:2:0.20	$CuCl_2$	Homogenous mixture	CuDES0.20	CuDES0.20 U
1:2:0.35	$CuCl_2$	Homogenous mixture	CuDES0.35	CuDES0.35 U
1:2:0.50	$CuCl_2$	Homogenous mixture	CuDES0.50	CuDES0.50 U
1:2:0.05	$SnCl_2$	Homogenous mixture	SnDES0.05	SnDES0.05 U
1:2:0.20	$SnCl_2$	Precipitation on cooling to 25 °C	_	_
1:2:0.35	$SnCl_2$	Precipitation on cooling to 25 °C	_	_
1:2:0.50	$SnCl_2$	Precipitation on cooling to 25 °C	_	_
1:2:0.05	$ZnCl_2$	Homogenous mixture	ZnDES0.05	ZnDES0.05 U
1:2:0.20	$ZnCl_2$	Precipitation on cooling to 25 °C	_	_
1:2:0.35	$ZnCl_2$	Precipitation on cooling to 25 °C	_	_
1:2:0.50	$ZnCl_2$	Precipitation on cooling to 25 °C	_	_

2.2. Methods

2.2.1. Formulation of MCDES

The reagents utilized in DESs formulation, viz. ChCl, glycerol and the metal chlorides are hygroscopic in nature therefore each of these were dried in a vacuum dryer for 10 h at 70 °C and 400 mmHg pressure prior to preparation. A basic DES was prepared by combining choline chloride and glycerol in a 1:2 M ratio and this mixture was stirred for 3 h at 80 °C resulting in the formation of a clear liquid. To this DES, varying concentrations of metal chlorides were added and stirring was continued at the same temperature for further 2 h until a clear homogenous solution was obtained. The composition of the formulated MCDESs are presented in Table 1. Although higher concentration of zinc chloride and tin chloride formed homogenous mixtures initially, but upon being cooled (25 °C), resulted in precipitation, thus these mixtures were not utilized for further investigation. The precipitation in these cases might be due to the lower solubility of these metal salts in the choline chloride based DES.

2.2.2. Characterization of the formulated MCDESs

The MCDESs were characterized using FTIR, rheology and interfacial surface tension.

FTIR characterization was performed using Spectrum 100 FTIR instrument of Perkin Elmer with a wavelength range of 4000–450 cm^{-1} to determine the bond structure in MCDESs as well as the alteration in different molecular bonding ensuing desulfurization. In order to perform FTIR of the liquid samples, these were immobilized on IR grade KBr and pelletized.

Rotational viscometer ViscoQC 300-L of Anton Parr GmbH, Austria equipped with the bob and cup arrangement (DG26 spindle) was used to determine the dynamic viscosities of the MCDESs and the model jet fuel at a temperature of 25 °C.

The interfacial surface tension of the different MCDESs and model fuel interface was measured using the Du Nuoy, Testmaster tensiometer of Testing Instruments at a temperature of 25 °C.

The Kamlet–Taft solvatochromic (K-T) parameters of the MCDESs were determined from the maxima UV–vis absorption peaks of the solutions of three dyes (*N*,*N*-diethyl-4-nitroaniline, 4-nitroaniline, and Nile red). These parameters were used to quantify the hydrogen-bond donating ability (α), the hydrogen bond accepting ability (β) and polarity/polarizability (π^*) of MCDESs utilizing the respective dyes. A high β value might have a direct effect on the desulfurization capacity of the DES through the formation of hydrogen bonding with the lone pair of sulfur. While a high π^* indicates better stability of the DES liquid which could enable better recycling capacity [22–25]. Eqs. 1–4 were used to evaluate these parameters.

$$\upsilon(dye) = \frac{10^4}{\lambda_{max,dye}} \tag{1}$$

$$\pi^* = 0.314 \times (27.520 - \upsilon_1) \tag{2}$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - \upsilon_2}{1.6078} \tag{3}$$

$$\beta = \frac{1.035 \times \upsilon_1 + 2.64 - \upsilon_2}{2.8} \tag{4}$$

Here, υ_1, υ_2, υ_3 are the values of absorbance for N,N-diethyl-4-nitroaniline, Nile red and 4-nitroaniline respectively.

Bruker 400 MHz ^{H}NMR was used to characterize the CuDES0.5. Deuterated DMSO was used as the solvent in this investigation to analyse the structure of DES.

The biodegradability of MCDES was evaluated from the values of Chemical Oxygen Demand (COD) Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD) using eqs. 5–8. In order to access the COD of sample water (pond water), 50 ppm of MCDES (CuDES0.5) was incorporated in 250 ml of sample water along with the addition of small amount of $HgSO_4$,1 ml Ag_2SO_4 in concentrated H_2SO_4, 10 ml of standard $K_2Cr_2O_7$ (0.05 N). This mixture was then refluxed at 100 °C followed by titration against standardized Mohr's salt solution (0.025 N) using ferroin indicator. COD was also evaluated for the sample water without MCDES to gauge its blank value. The DO of the water sample was measured by sampling out 10 ml (from solution containing MCDES as well as blank) from BOD bottle and adding 0.08 ml $MnSO_4$ (2 M) 0.08 ml of standard alkali iodide azide and 1 ml H_3PO_4. This mixture was titrated against standard $Na_2S_2O_3$ (N/20) using starch indicator. The BOD test was performed by taking 250 ml of sample water inoculated with 50 ppm MCDES and adding phosphate buffer, $MgSO_4$ (saturated solution), $CaCl_2$ (saturated solution), $FeCl_3$ (saturated solution) each of 1 ml. The contents of the bottle was aerated and kept in shaker incubator at 25 °C. The BOD was calculated from the DO values at 5 day intervals [26,27].

$$Amount\ of\ DO\ (in\ mg) = \frac{100 \times V \times S}{V_1 - V_2} \tag{5}$$

where S = Strength of $Na_2S_2O_3$ solution; V = Net volume of water in ml; V_1 = Volume of (water sample + reagent solution); V_2 = Volume of ($MnSO_4$ + reagent solution)

$$BOD\left(in\frac{mg}{l}\right) = [(D_0 - D_5) - (C_0 - C_5)] \times dilution\ factor \tag{6}$$

where D_0 = Initial value of DO; D_5 = DO after 5 days incubation at 25 °C; C_0 = Initial value.

of DO of dilution water; C_5 = DO after 5 days incubation at 25 °C of dilution water

Table 2
Independent variables and their levels used for Central Composite Design (CCD).

Independent variables	Coded levels	
	-1	+1
Solvent to feed ratio (A)	0.1	0.6
Concentration of metal chloride in MCDES (mole ratio) (B)	0.05	0.5
Extraction time (min.) (C)	30	180

$$Amount\ of\ COD\left(in\frac{mg}{l}\right) = 20 \times 8 \times (V_2 - V_1) \times S \quad (7)$$

where S = Strength of $K_2Cr_2O_7$ solution; V_1 = Titre volume of sample water; V_2 = Titre volume in blank titration

$$\%Biodegradation = \frac{BOD}{COD} \times 100 \quad (8)$$

2.2.3. Molecular dynamic simulation

Geometry optimization of the MCDESs and model jet fuel components were performed by the $DMol_3$ module of Materials Studio5.5.2. The optimization parameters which were utilized are, Meta-Generalized Gradient Approximation (m-GGA) and Molecular Out-of-distribution Generalization-1 (MOG-l) functionals along with Double Zeta with Polarization (DNP) as the basis set. The maximum force, maximum displacement, maximum iteration, maximum step size, SCF tolerance, maximum SCF cycle, multipolar expansion, spin, were 0.004Ha/Å, 0.005 Å, 1000, 0.3 Å, 0.2, 1000, hexadecapole and 0.5 respectively. These calculations yielded the sigma profiles of the components of the solvents as well as the model fuel, which were analysed to get an insight of the desulfurization process [28–30].

2.2.4. Model jet fuel preparation and desulfurization process

The model jet fuel was prepared by combining n-decane and toluene in a volumetric ratio of 4:1 and the sulfur compounds (thiol and BT) were added to this mixture at a concentration of 5000 ppm on elemental sulfur basis. Each of the MCDESs was mixed with the model jet fuel in the ratio of 1:5 (*v*/v) and agitated using magnetic stirrer at 400 rpm maintaining a temperature of 25 °C. Samples of the treated model fuel were collected at the end of 3 h. The sulfur content after desulfurization was analysed using Trace 1300 Gas Chromatograph (GC) by Thermo Fischer Scientific Inc., USA equipped with TR-5 GC capillary column having an internal diameter of 0.32 mm, length of 30 m and a film thickness of 1×10^{-2} mm and a flame ionization detector (FID). The flow rate of carrier gas (nitrogen) was maintained at 1.5 ml/min.

All the experiments were carried out in triplicate and the average value was reported.

The desulfurization efficiency was determined using the following eq. (9):

$$\text{Percentage desulfurization} = \frac{I - F}{I} \times 100\% \quad (9)$$

where, I denotes the initial concentration (ppm) on elemental sulfur basis.

F denotes the final concentration (ppm) on elemental sulfur basis.

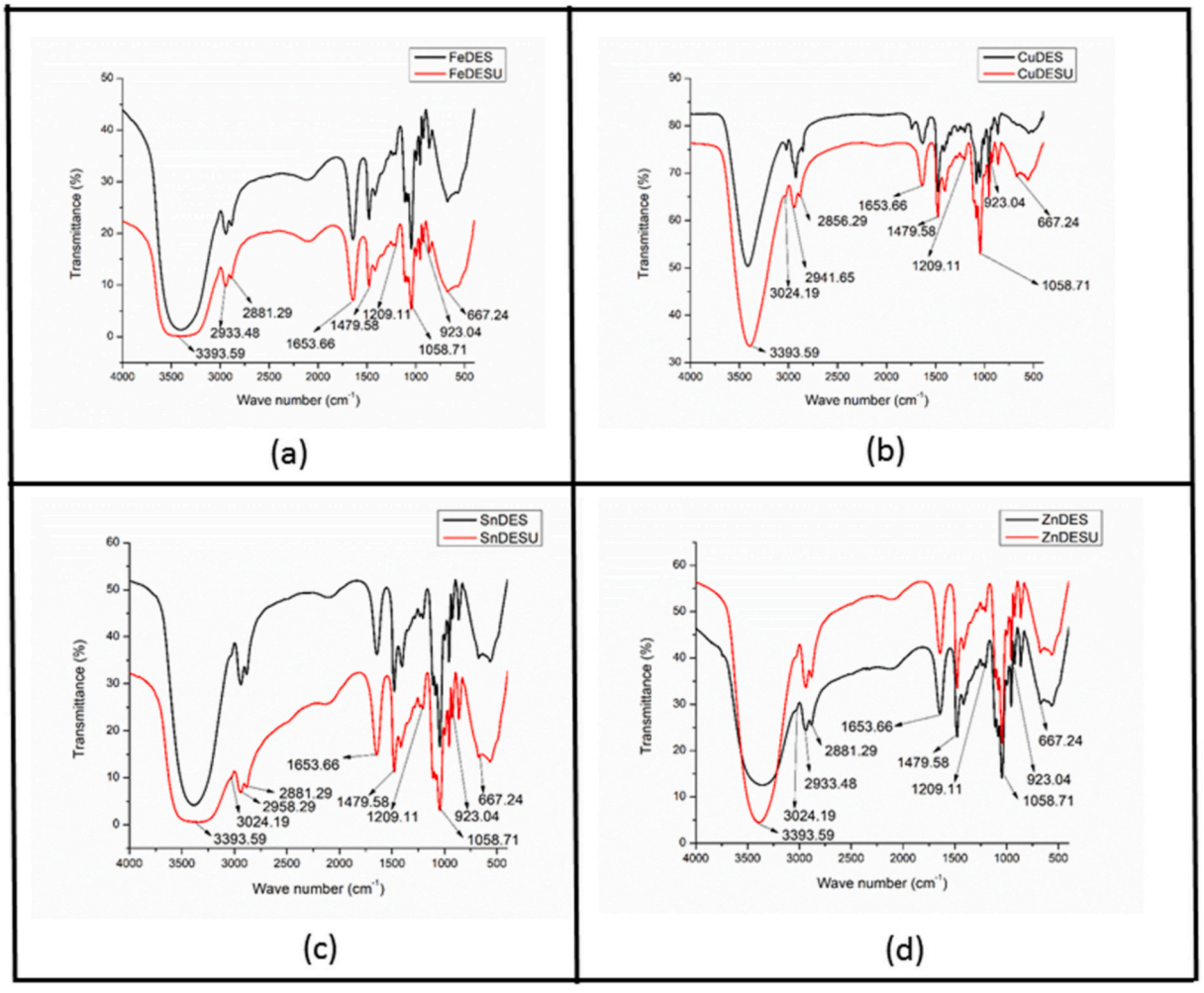


Fig. 1. FTIR spectra of different MCDESs before and after desulfurizing model jet fuel.

2.2.5. Optimization of desulfurization process

Design Expert 8.5 software was used for process optimization using the Analysis of Variance (ANOVA) for the accuracy evaluation of the liquid-liquid extraction process. A three-factor-two-level Central Composite Design (CCD) was used to optimize the extraction conditions. Based on the experimental factors solvent to feed ratio (A), concentration of metal chloride in MCDES (mole ratio) (B), and extraction time (min.) (C) were chosen as independent variables in CCD and tested in a 20-run experiments (with 7 central point runs). Each independent variable was coded at two levels, −1 and + 1, for low and high levels, respectively and is represented in Table 2 [31].

2.2.6. Error analysis

The error analysis is required for evaluating the accuracy the liquid-liquid extraction process. The error range (approximately 0.05 %) for the experiments performed in triplicates are depicted as error bars in the respective plots [32,33].

2.2.7. Regeneration and reusability of solvent

The used MCDES (CuDES0.5) was regenerated by dissolving in water to separate out the absorbed sulfur compounds as raffinate (top layer). The aqueous solution of MCDES was then collected and the water was removed by rotary vacuum evaporation at a temperature of 70 °C [4].

3. Results and discussions

3.1. FTIR characterization

Fig. 1 represents the FTIR of the MCDESs (FeDES0.05, CuDES0.05, SnDES0.05 and ZnDES0.05) before and after desulfurization. The peak at 3393.59 cm^{-1} is attributed to the asymmetric N—H stretching of ChCl [29].This peak might have overlapped the O—H stretching peak for glycerol which lies at 3265 cm^{-1} [12]. The O—H bond stretching of ChCl is depicted by the peaks lying in the range of 3000-2900 cm^{-1} [1,34]. Another peak pertaining to the C—H stretching in glycerol is present in the wave number range of 2943-2873 cm^{-1}in all the FTIR spectra [34]. The presence of the peaks of glycerol and ChCl in the unused MCDESs signify that no reaction has occurred between the HBA and HBD of the unused MCDESs. The peak at 1653.66 cm^{-1} which corresponds to the C═O stretching vibration might be attributed to the presence of trace amounts of esters in the glycerol used for DES formulation [35,36]. Another peak pertaining to the CH_3 and CH_2 rocking vibration in ChCl at 1479.58 cm^{-1} is also observed in all the spectra for the unused solvent [37]. The C—O stretching vibration for glycerol is depicted by the peak at 1209.11 cm^{-1} [35]. The C—N stretching vibration of ChCl is indicated by the peak at 1058 cm^{-1} [38]. The peak at 923 cm^{-1} is assigned to the symmetric C—N stretching of ChCl [39].

The peaks lying in the range of 750-450 cm^{-1}exhibit the presence of metal chlorides in the MCDESs. These peaks for the metal chlorides are attributed to the M-Cl bond stretching vibrations [1,40].

The peak widening at 3393.59 cm^{-1} after desulfurization in case of all the solvents indicate that desulfurization mainly takes place through hydrogen bonding between the sulfur impurities and the components of the MCDESs [20]. The peak at 3024.19 cm^{-1} pertaining to the CH_3 out of plane stretching of thiol is the most prominent in case of used CuDES which signifies that this particular solvent might be most selective in removing thiol [41]. It is also noted that in the case of CuDESU (Fig. 1 (b)) there was a significant elongation in the peak at 1653.66 cm^{-1} which suggested that the Cu^{2+} ion in CuDES can form π-π complexion with the aromatic ring of BT [20].The elongation of the peaks at 1479.58 cm^{-1}denotes the C═C stretching of aromatic ring probably due to the presence of BT in the used MCDESs [42]. The peak values lying within 1276.91 to 1209.08 cm^{-1} that correspond to the bending atomic motions of the thiophenic ring of BT, also exhibit the existence of this compound in the used MCDESs [43].

Table 3
Dynamic viscosities of model jet fuel and MCDESs:

MCDESs	Dynamic viscosity (mPa.s)
Model jet fuel	0.568
FeDES0.05	213.740
FeDES0.20	668.280
FeDES0.35	1218.320
FeDES0.50	2340.130
CuDES0.05	363.700
CuDES0.20	1594.320
CuDES0.35	896.100
CuDES0.50	978.200
SnDES0.05	339.960
ZnDES0.05	423.700

Table 4
Interfacial surface tension between model jet fuel and MCDESs phases.

MCDESs	Interfacial surface tension (mN/m)
FeDES0.05	48.0
FeDES0.20	46.0
FeDES0.35	46.8
FeDES0.50	49.9
CuDES0.05	37.6
CuDES0.20	45.1
CuDES0.35	47.1
CuDES0.50	52.5
SnDES0.05	34.1
ZnDES0.05	29.6

Table 5
K-T parameters of the MCDESs.

MCDES	α	β	Π*
FeDES0.05	1.51	0.53	1.12
FeDES0.2	1.58	0.67	1.18
FeDES0.35	1.57	0.61	1.24
FeDES0.5	1.69	0.74	1.27
CuDES0.05	1.53	0.55	1.16
CuDES0.2	1.59	0.69	1.25
CuDES0.35	1.66	0.70	1.32
CuDES0.5	1.72	0.79	1.38
SnDES0.05	1.52	0.54	1.13
ZnDES0.05	1.49	0.52	1.11

3.2. Rheology analysis of the MCDESs

The Table 3 lists the viscosities of the MCDESs and model fuel. For the MCDESs, it is observed that the values of viscosities are in the order of 10^3mPa.s while that of the model fuel is 10^{-1} mPa.s. Thus the viscosity difference between MCDESs and the model fuel are in the magnitude of 10^4mPa.s which indicate that the extract and raffinate phases are expected to separate out quickly [44,45].

3.3. Determination of the surface tension of the model fuel-MCDES interface

In a liquid-liquid extraction system the interfacial surface tension plays a crucial role in mass transfer between the two phases and a low surface tension is preferable. However, a very low interfacial surface tension is not desirable since this condition requires longer time for phase separation at the end of the extraction cycle and can also lead to emulsification [46]. It is observed that the values of interfacial surface tension between the MCDESs and model jet fuel phases lie in the order of 10^{-2} N/m, which imply that the dispersed phases will coalesce rapidly (within 5 min) thereby keeping the equilibrium holdup time to a minimum [44]. The values are represented in Table 4.

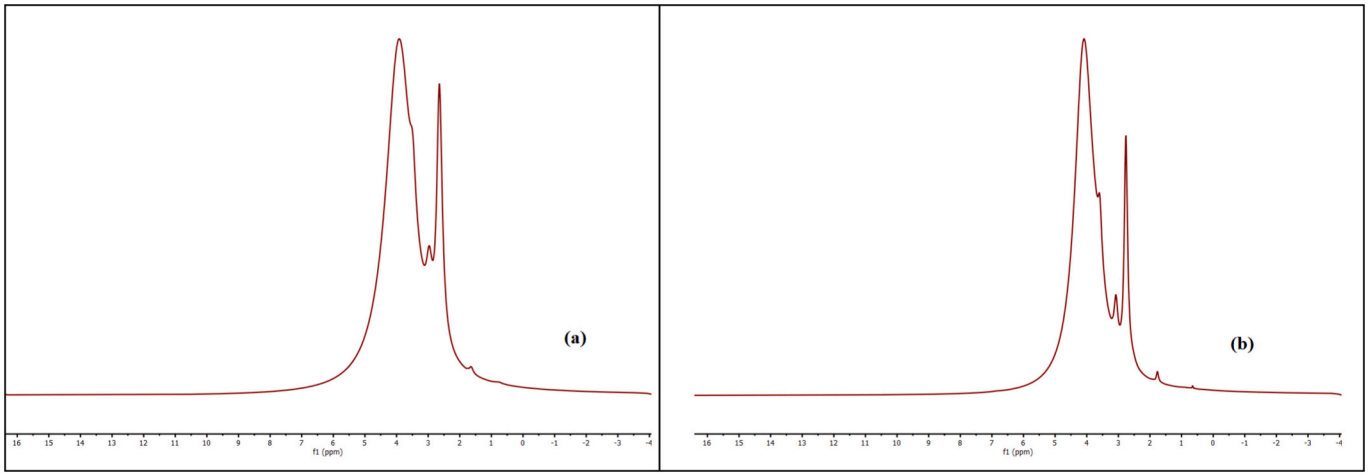


Fig. 2. (a) ^{H}NMR spectra of CuDES0.5 (b) ^{H}NMR spectra of CuDES0.5 U.

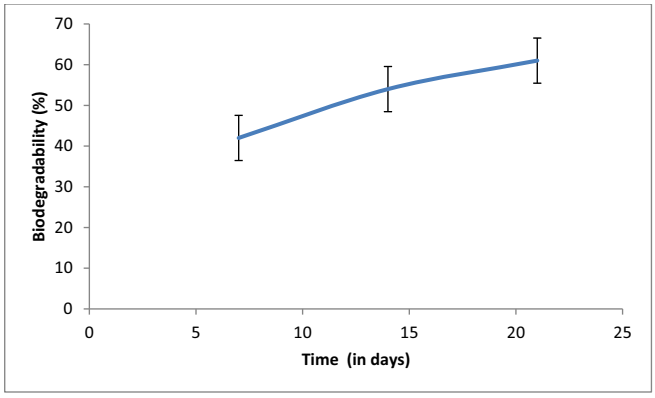


Fig. 3. Biodegradation behaviour of CuDES0.5.

3.4. Estimation of Kamlet–Taft solvatochromic parameters of the MCDESs

The Kamlet-Taft solvochromic parameters pertaining to HBD ability (α), HBA ability (β) and polarizability (π*) of the MCDESs are reported in Table 5. It is observed that α and β values of CuDES0.5 are the highest which indicates best solvent stability. The greatest value of β of this MCDES also signifies that it might form good hydrogen bonding with the lone pair of sulfur molecule which is expected to enhance the desulfurization efficiency. It is also observed that with the increase in metal chloride concentration ($FeCl_3$ and $CuCl_2$) π* increases, hinting at the enhanced stability of the solvents. The high polarizability of the solvents is expected to enhance their water solubility during their regeneration process [22,23,25,47].

3.5. NMR characterization

Since CuDES0.5 exhibited the highest desulfurization efficiency as reported in section 3.8, this as well as the used solvent (CuDES0.5 U) were characterized using ^{H}NMR analysis (Fig. 2).

The peaks at 2.64 and 2.26 ppm for CuDES0.5 and CuDES0.5 U represents duterated DMSO which was used as the solvent for ^{H}NMR analysis. ChCl exhibits its characteristic peaks at 4.02 ppm and 3.56 ppm which have shifted to 3.92 ppm and 3.48 ppm respectively which could be attributed to the hydrogen bond interactions in CuDES0.5 [48,49]. While the peak at 1.40 ppm in CuDES0.5 might hint to the presence of methyl ester impurity in glycerol as had been deduced from the FTIR analysis [50]. The distinctive peaks for glycerol at 3.68 ppm and 3.59 ppm might have been overlapped by the peaks of ChCl [48]. The appearance of peak at 0.65 ppm as well as the increase in peak intensity at 1.76 ppm (originally at 1.40 ppm for CuDES0.5) might be attributed to the incorporation of the sulfur compounds in the used MCDES (CuDES0.5 U). The appearance of peak at 4.90 ppm for CuDES0.5 U could hint to the breaking of hydrogen bond in the MCDES after desulfurization [51].

3.6. Estimation of biodegradability of MCDES

Fig. 3. depicts the preliminary biodegradability characteristics of the CuDES0.5 against days. The biodegradation characteristic of CuDES0.5 was evaluated since this solvent showed the highest desulfurization efficiency as reported in section 3.8. It is observed that after 21 days, the % biodegradation of CuDES0.5 was 61.02 %, which indicates good biodegradable behaviour, thereby making it an environment friendly solvent. The high biodegradability of this solvent might be attributed to its highly biodegradable precursors i.e. choline chloride and glycerol [26,52].

3.7. Estimation of MCDESs desulfurization efficiency from sigma profile

Sigma profile of thiol, BT and the individual MCDESs components are depicted in Fig. 4. In a sigma profile, the hydrogen bond accepting (HBA) ability of compounds is shown by the peaks that lie beyond the screening charge density value of −0.0082, whereas the hydrogen bond donating (HBA) ability is indicated by the peaks that lie beyond +0.0082. The non-polar region is indicated by the space between these two screening charge density values, or between −0.0082 to +0.0082. Weak hydrogen bonding is exhibited by compounds that have peaks in the non-polar area [8,28].

From the sigma profiles depicted in Fig. 4 it is observed that BT has two major peaks. The peak of BT in the negative coordinate is assigned to the electropositive positive carbon atoms adjacent to the sulfur atom, while the large peak in the positive coordinate is due to the negative charge on its sulfur atom. It is also observed that the major peaks of BT are distributed very close to the zero screening charge density which suggests its strong non-polar characteristics. Therefore, from the 'like dissolves like' rule, it can be inferred that BT will be preferentially soluble in non-polar solvents [53]. In case of thiol, it is observed that a major peak is present in the negative side of the non-bonding region, signifying that it is also non-polar in nature. However, a small part of its sigma profile extends into the HBD region hinting at its weak hydrogen bonding ability [8].On analyzing the screening charge density distributions of the components of MCDESs, it is evident that glycerol is nonpolar in nature, since its peaks are entirely distributed in the non-

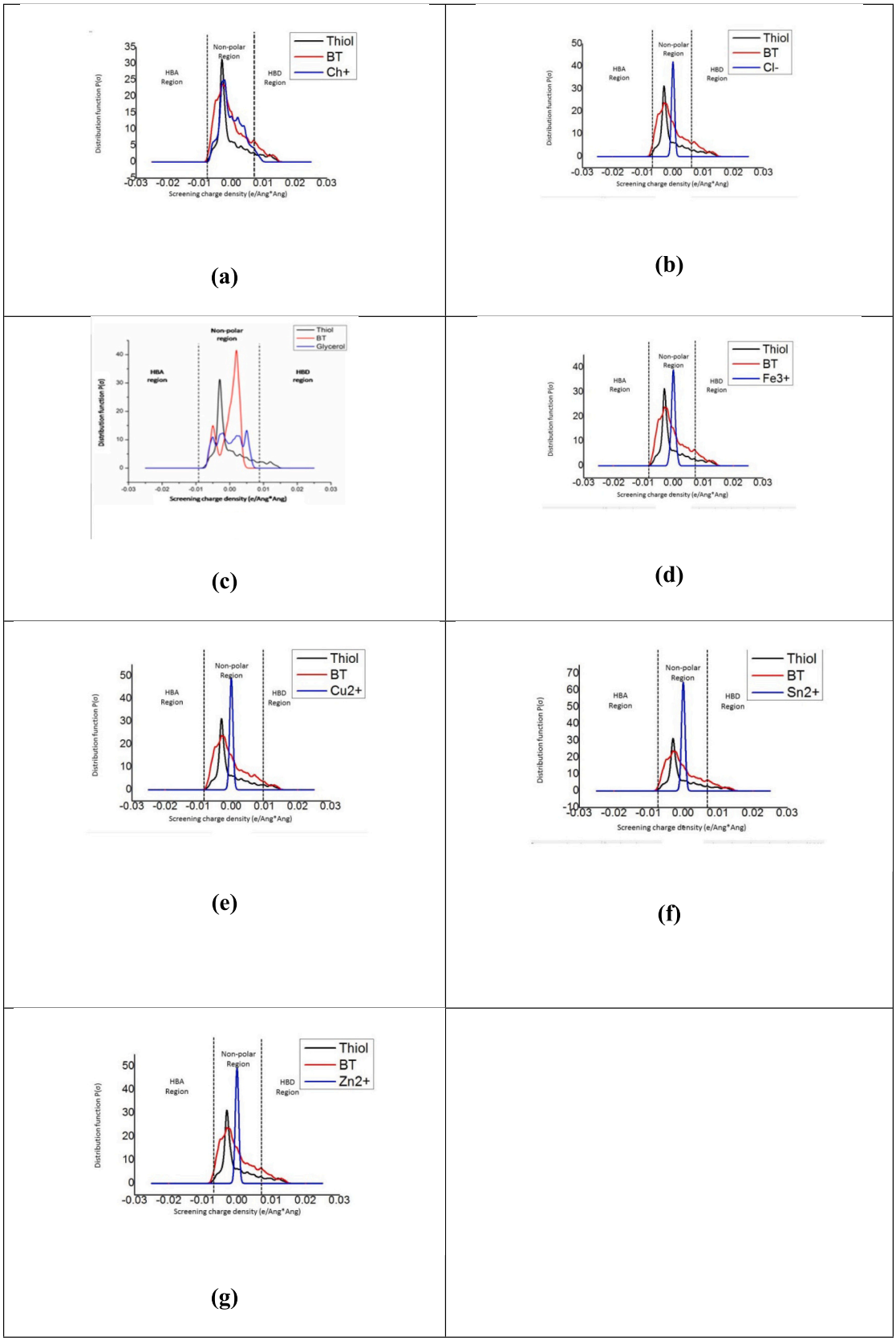


Fig. 4. Sigma profile of thiol, BT and the MCDESs components.

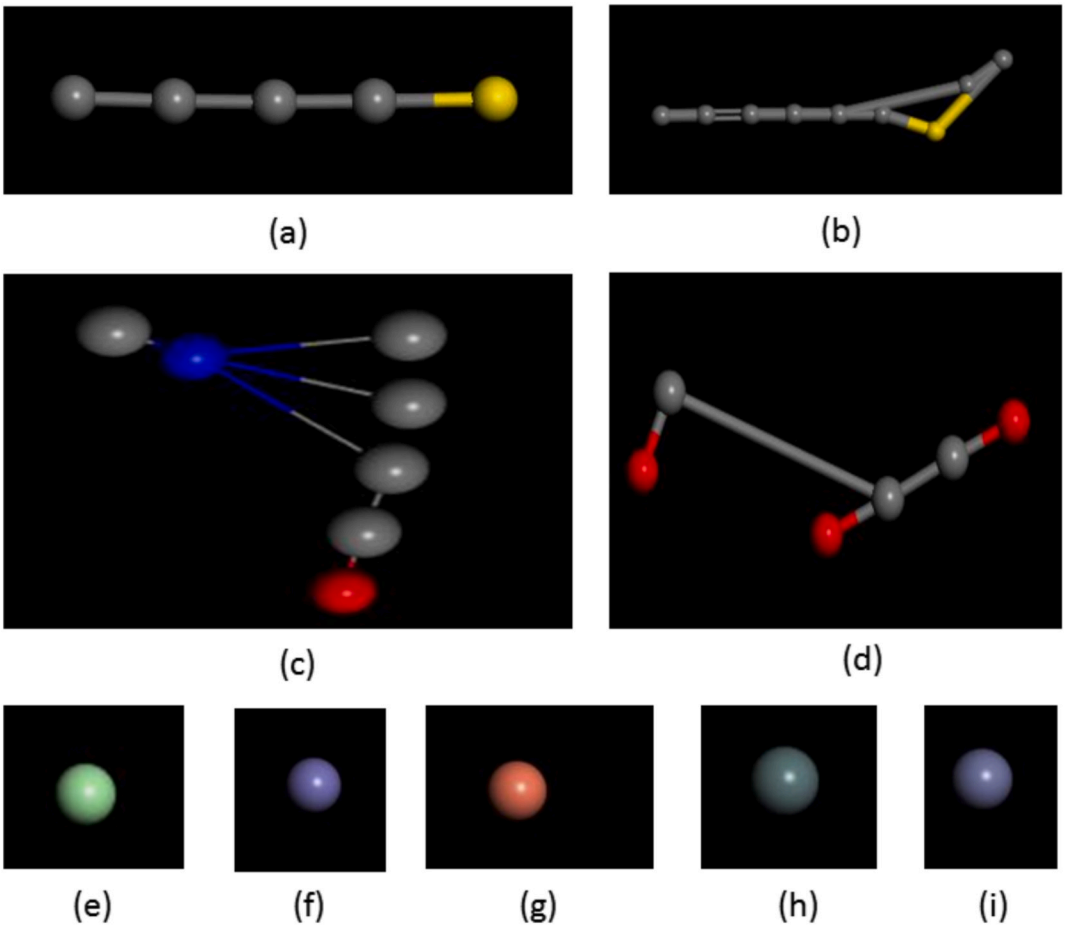


Fig. 5. Geometry optimized structures of (a) 1-butane thiol (b) Benzothiophene (c) Choline ion (d) Glycerol (e) Chloride ion(f) Iron ion(g) Copper ion (h) Tin ion (i) Zinc ion.

bonding region. However, in case of choline ion, it is seen that its sigma profile nearly overlaps with those of the sulfur compounds under consideration (BT and thiol), which implies that a choline based solvent will be highly selective towards these compounds. Additionally, the peak of the chloride ion lies close to the zero screening charge density and nearly mirrors the peak of thiol signifying that chloride ion can interact with this compound, thereby increasing its solubility. Overlapping of the peaks of the chloride ion and that of BT, indicate that mutual interaction is possible, that might yield enhanced solubility of BT in HBA [8,53,54].

The investigation of the sigma profiles of the metal ions reveal that, all of these peaks exist in the zero screening charge density region. This

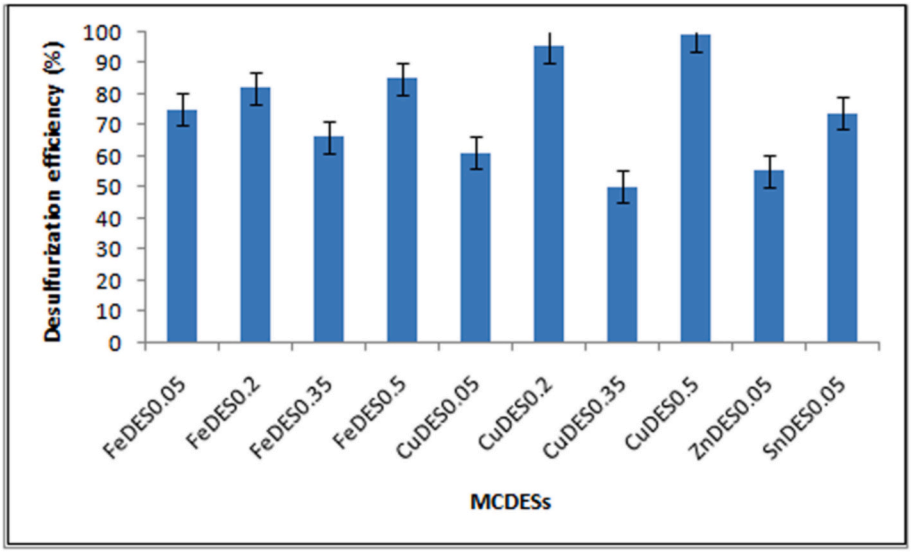


Fig. 6. Desulfurization efficiency of the different MCDESs for the model jet fuel after 3 h.

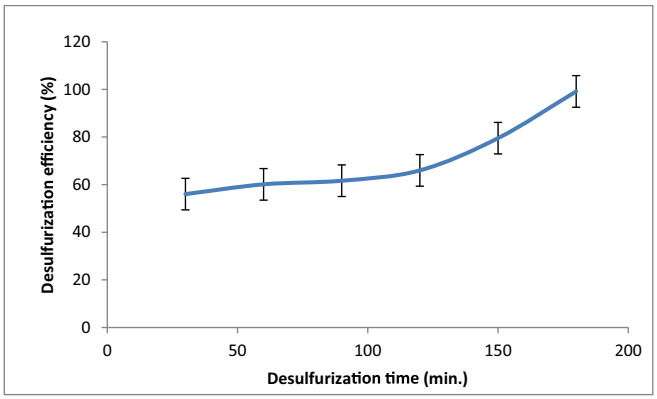


Fig. 7. Desulfurization efficiency of model jet fuel using CuDES0.5.

shows that incorporation of these metal ions in the DES can enhance the desulfurization efficiency towards thiol since the peaks of these metal ions mirror the major peak of thiol indicating their mutual solubility. The peaks of the metal ions also share a common region with that of BT which also manifests their mutual solubility. Among these metal ions, it is noted that the Cu^{2+} and Sn^{2+} have larger elongated peaks thereby denoting higher distribution function that might contribute to better interaction with the corresponding sulfur compounds, resulting in enhanced solubility [8,53].

The optimized molecular structures of the MCDESs components and model sulfur compounds as obtained from the $DMol_3$ module of Materials Studio 5.5.2 are represented in Fig. 5. The bond lengths of the different components of choline ion as obtained from the optimized structures are N-C(methyl) ~1.53 Å, N-C(side chain) ~1.60 Å, C-C ~ 1.56 Å, and C-O ~ 1.39 Å. However, the bond lengths in the optimized structure of glycerol are 1.39 Å C-O ~ 1.39 Å and O-H ~ 1.1 Å and C-C ~ 1.54 Å.

3.8. Desulfurization performance analysis of the MCDESs

Fig. 6. depicts the desulfurization performance of the MCDESs towards the model jet fuel after a treatment time of 3 h. From the plot it is evident that for the metal chlorides used, desulfurization performance follows the trend $CuCl_2$ > $FeCl_3$ > $SnCl_2$ > $ZnCl_2$. This result is in accordance with the sigma profile analysis of the MCDESs. The higher hydrogen bond forming tendency of CuDES with both thiol and BT, compared to the other MCDESs as is evident from the FTIR of the used MCDESs implies that more sulfur compounds get extracted by this solvent. It is also observed that the desulfurization tendency of copper based DESs varies as CuDES0.5 > CuDES0.35 > CuDES0.2 > CuDES0.05. This could be attributed to the high interaction capacity of the Cu^{2+} ion with the respective sulfur compounds as inferred from the molecular dynamic investigation. It is also evident from the β value of K-T parameters that CuDES0.5 possess the highest HBA ability which might be responsible for the best desulfurization performance by this MCDES. On the other hand in case of the iron based DES, the trend of desulfurization is as follows FeDES0.5 > FeDES0.2 > FeDES0.05 > FeDES0.35.

The extracted sulfur compounds might be used as industrial raw materials or treated with caustic solution to produce sulphites and hydrocarbons thereby converting them into value added products [4].

A further desulfurization investigation on the basis of total sulfur in the model jet has been carried out in the case of CuDES0.5, to get an insight into the desulfurization process which is depicted in Fig. 7. The desulfurization capacity of this particular solvent is 56.02 % after 30 min of extraction which steadily increases to 99.15 % at the end of 3 h. However, since the sulfur content specification for JetA-1 fuel (84.90 ppm is much less than 3000 ppm) has already been met, further extraction was discontinued.

Table 6
Design matrix and response of the desulfurization process.

Run no.	Solvent to feed ratio Factor (A)	Concentration of metal chloride mole in MCDES (mole ratio) Factor (B)	Time of extraction (min.) Factor (C)	Desulfurization efficiency (%)
1	0.6	0.09	180.00	68.3
2	0.38	0.22	113.00	76.0
3	0.38	0.22	113.00	76.0
4	0.60	0.35	180.00	78.9
5	0.38	0.22	225.68	98.2
6	0.38	0.00	113.00	56.7
7	0.16	0.35	180.00	50.0
8	0.16	0.35	46.00	62.3
9	0.16	0.09	46.00	15.8
10	0.38	0.22	113.00	76.0
11	0.38	0.22	113.00	76.0
12	0.38	0.22	0.32	12.9
13	0.01	0.22	113.00	20.0
14	0.16	0.35	46.00	61.4
15	0.60	0.09	46.00	15.2
16	0.38	0.44	113.00	68.9
17	0.38	0.22	113.00	76.0
18	0.38	0.22	113.00	76.0
19	0.75	0.22	113.00	76.0
20	0.16	0.09	180.00	25.1

3.9. Extractive desulfurization process optimization by ANOVA

In this work, the statistical design software Design Expert (Stat-Ease Inc., Minneapolis, USA) – DX7 was used to analyse the data. Central composite design was implemented for the experimental design. This design consists of selecting three experimental points (cube points) that come from the factorial design in addition to axial and center points. In this work, a total of 20 experiments were used for each model in the form of one block with seven center points. The parameters for copper chloride containing MCDESs were optimized by this method since these exhibited higher desulfurization efficiency in comparison to the other metal chloride based MCDESs which were selected in this investigation. Extraction time, concentration of metal chloride in the MCDES and solvent to feed ratio, were initially considered as the factors affecting the extraction efficiency in this investigation. The solvent to feed ratio, the concentration of metal chloride in the MCDES and the extraction time were varied from 0.16 to 0.6, 0.09 to 0.35 (mole ratio) and 46 to 180 min respectively. The design matrix and the responses are given in Table 6.

The final empirical model equation for the desulfurization efficiency using the CuDES0.5 is as follows:

$$\text{Extraction efficiency} = 7.629 + 16.65^{*}A + 3.36^{*}B + 25.30^{*}C - 3.54^{*}A^{*}B + 10.74^{*}A^{*}C - 5.34^{*}B^{*}C - 11.82^{*}A^{2} - 6.58^{*}B^{2} - 9.18^{*}C^{2} - 3.76^{*}A^{*}B^{*}C + 14.46^{*}A^{2^{*}}B - 18.64^{*}A^{2^{*}}C - 6.69^{*}A^{*}B^{2} \quad (10)$$

Here, the CuDES extraction efficiency is represented as a function of the input variables (A,B,C). The positive and negative signs before the terms show the respective synergetic and antagonistic effects of their variables [55].

Analysis of variance (ANOVA) was used to study the reliability of the sulfur removal model. Table 6 gives the model fitness parameters of the CuDES0.5. The model *p*-value of 0.0004 suggests that the model is highly significant for predicting the experimental data. The prediction coefficient of the determinant (R^2) 0.9813 for desulfurization efficiency is in reasonable agreement with the adjusted R^2 of 0.9408 exhibiting a good predictability of the model. The Adequate Precision which measures the signal to noise ratio, should be greater than 4. In this work, Adequate Precision was found to be 15.353 indicating adequate signal quality [55]. Table 7 depicts that the "Lack of Fit p-value" of the model

Table 7
Model fitness parameters for CuMCDES.

Model parameter	Extraction efficiency values
R^2	0.9813
Adjusted R^2	0.9408
Adequate precision	15.353
Std. Dev.	6.63
Mean	58.05
P-value	0.0004
Lack of fit P-value	74.00
A (F-value)	35.72
B (F-value)	1.07
C (F-value)	82.50

for desulfurization is not significant, which confirms good model efficiency for prediction of extraction efficiency. The F values evaluated from ANOVA analysis indicate that the optimum values of solvent to feed ratio, concentration of metal chloride in MCDES and extraction time are 35.72, 1.07 mol ratio and 82.50 min respectively [55].

3.10. Regeneration and reusability of CuDES0.5

The regeneration of a solvent after extraction is vital from the industrial point of view, considering the huge amount of solvent. In this investigation, CuDES0.5 was regenerated and reused since it had exhibited the highest desulfurization performance among all the MCDESs. Fig. 8. depicts the regeneration performance of CuDES0.5. The desulfurization performance of the regenerated CuDES0.5 decreased from 99.15 % to 91.34 % after the first regeneration cycle and varied only slightly in the later recycles. The solvent was regenerated and reused 5 times. At the end of the fifth cycle, the desulfurization efficiency was 91.21 %. Thus it can be inferred that this solvent has the potential for reusability.

4. Conclusions

In this present investigation, four metal chlorides incorporated DESs have been formulated and their efficiency towards model jet fuel desulfurization was studied. The characterization of CuDES0.5 by ^{H}NMR indicated the presence of good hydrogen bonding in this MCDES validating the formation of DES structure. The FTIR analysis of MCDESs revealed that the desulfurization phenomenon occurred through hydrogen bond formation between the MCDESs and the sulfur compounds as well as by π-πcomplexion of the sulfur compounds with the metal ions. The effect of hydrogen bond on desulfurization performance was also revealed from the K-T parameters which indicated that CuDES 0.5 has the greatest HBA ability, which can be attributed to its highest desulfurization efficiency. Further characterization of the MCDESs by interfacial surface tension and rheology implied that the extract and raffinate phases are expected to separate out quickly minimizing the holdup time for phase separation. Additionally, the comparative analysis of the sigma profiles of the model sulfur compounds as well as the components of the MCDESs as obtained from the $DMol_3$ module of Materials Studio 5.5.2 has indicated that the CuDES has higher interaction ability with the sulfur compounds under consideration resulting in higher desulfurization efficiency. Model jet fuel desulfurization analysis was performed with the MCDESs at a temperature of 25 °C and the results were in accordance to the predictions from FTIR and molecular dynamic simulation indicating that, CuDES has the highest separation efficiency. Among the CuDESs prepared, CuDES0.50 (ChCl: glycerol:$CuCl_2$ molar ratio of 1:2:0.50) yielded the highest desulfurization efficiency (99.15 % on total sulfur basis) for a solvent:feed ratio of 1:5 and an extraction time of 3 h. The process of desulfurization was optimized by ANOVA which indicated that the optimum values of solvent: feed ratio, concentration of $CuCl_2$ in MCDES and extraction time are 35.72, 1.07 (mole ratio) and 82.50 min respectively. Again, in this investigation, the low amount of MCDESs utilized (1:5 volumetric ratio of solvent:model jet fuel) for this extractive desulfurization along with good regeneration ability of CuDES0.5 can contribute towards the economic viability of this process. The solvent CuDES0.5 also showed good biodegradability behaviour which manifests that this is an eco-friendly solvent.

CRediT authorship contribution statement

RupamChatterjee: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Debarati Mitra:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

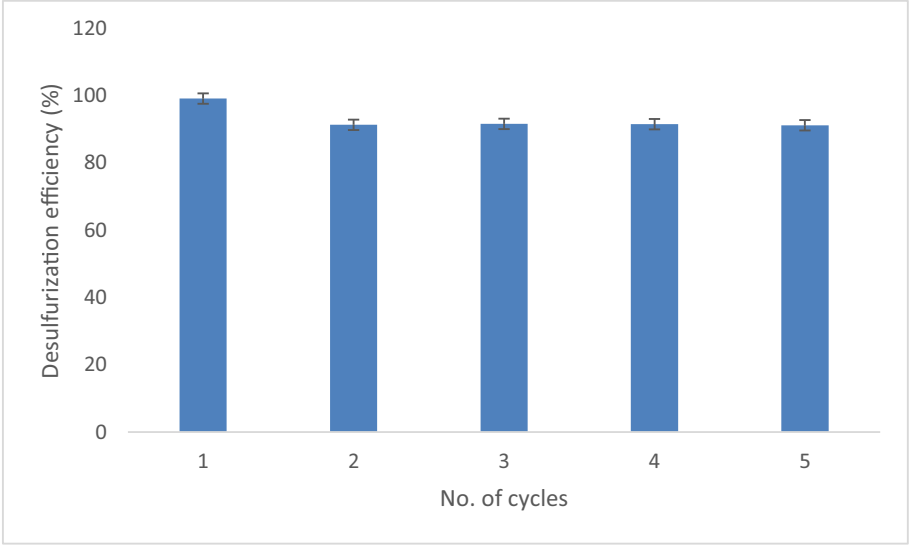


Fig. 8. Effect of regeneration of CuDES0.5 on desulfurization performance.

Data availability

The data that has been used is confidential.

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